

Experiment 13: Maximum Subarray Sum (Kadane's Algorithm)

Aim

To implement **Kadane's Algorithm** in Python to find the **maximum subarray sum** in a given array.

Algorithm

Step 1: Start.

Step 2: Read the number of elements n.

Step 3: Read the array elements.

Step 4: Initialize max_sum and current_sum with the first element.

Step 5: Traverse the remaining elements:

- Update current_sum as the maximum of the current element and current_sum + current element.
- Update max_sum if current_sum is greater.

Step 6: Display the maximum subarray sum.

Step 7: Stop.

Source Code (Python)

```
main.py    Share  Run
```

```
1 # Maximum Subarray Sum using Kadane's Algorithm
2
3 n = int(input("Enter number of elements: "))
4
5 arr = []
6
7 print("Enter the elements:")
8 for i in range(n):
9     arr.append(int(input()))
10
11 max_sum = arr[0]
12 current_sum = arr[0]
13
14 for i in range(1, n):
15     current_sum = max(arr[i], current_sum + arr[i])
16     max_sum = max(max_sum, current_sum)
17
18 print("Maximum Subarray Sum =", max_sum)
```

Output

Output

Enter number of elements: 8

Enter the elements:

-2

-3

4

-1

-2

1

5

-3

Maximum Subarray Sum = 7

=== Code Execution Successful ===

Time Complexity

- Best Case: $O(n)$
- Average Case: $O(n)$
- Worst Case: $O(n)$

Space Complexity

- $O(1)$

Result

The Maximum Subarray Sum was successfully found using Kadane's Algorithm in Python. The program correctly identifies the contiguous subarray with the largest sum and displays the maximum sum.